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Noetic Topos 2.0: A Marked and Modal Successor to the Conservative Noetic Site

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Abstract

The conservative Noetic Topos construction contains a valid conditional topos-existence argument, but its ACBE and RPM cover clauses do not retain the separate information later interpretations assigned to them: ACBE reflexivity places an identity in the corresponding sieve, and the RPM counit — split by its section — forces an identity by sieve closure. Each clause alone generates only maximal sieves, and the combined topology collapses to the jointly-surjective coverage already present in the source. Distinct modalities cannot be recovered by relabeling those generators.

Noetic Topos 2.0 is the first conservative successor that survives this collapse counterexample. It augments the legacy site with a finite four-atom context lattice `L = P({0,1} × {0,1})` whose coordinates carry formal ACBE and RPM marks, equips `L` with two transverse coordinate-partition coverages, and

keeps immutable, hash-chained cover provenance *outside* the ordinary topology. On `PSh(L, Set)`, internal proofs backed by exhaustive finite checks establish that both generated topologies are subcanonical, their sheafification reflectors are distinct and non-identity, their join is the topology generated by the four-singleton partition, and the reflectors commute up to canonical natural isomorphism. The ambient candidate `NoeTop_2 = PSh(C × L)` carries three modal reflectors and a resolved subtopos; the full research object `NoeSystem_2` packages these with an external mark record. The product-site lift, legacy round-trip comparison, cross-context coupling, proof-assistant formalization of the sheaf layer, and independent category-theory review remain open, and the paper's claim discipline treats preserving those boundaries as part of the result.

---

1. The defect being repaired

Let `C` be the registered legacy category `Noetica` and `J_legacy` its proven coverage. The conservative source admits covering sieves from three routes: jointly-surjective observational families, designated ACBE descent maps, and the RPM counit. Two sieve arguments are decisive:

1. **ACBE.** The reflexive descent map at equal contexts is `id_X`. A sieve on `X` containing `id_X` is maximal.
2. **RPM.** The section-retraction data `ε_X ∘ η_X = id_X` mean that any sieve containing `ε_X` also contains `id_X` after closure under precomposition — again maximal.

Hence `J_legacy = J_jointly-surjective`: ordinary sieve generation has *forgotten* the distinct ACBE and RPM provenance. This does not invalidate the generic topos-existence statement; it narrows what the topology records. A successor must change the cover generators — renaming the old maps is not a repair.

A second, independent gap: there is no derivation of subcanonicity for the legacy topology. A finite countermodel (sets of size ≤ 2 with injections) exhibits a jointly-covering pair of point inclusions whose compatible local injections into a singleton admit no global injection. Until a revised site is proved subcanonical, the distinguished legacy object is represented in the sheaf topos by `a_legacy(yX)`, never by silently treating `yX` as a sheaf.

2. The finite marked context factor

2.1 Context lattice and transverse coverages

Define

```
Omega = {0,1} × {0,1},     L = P(Omega),     U -> V  iff  U ⊆ V.
```

The first coordinate is the formal ACBE mark; the second the formal RPM mark. These are structural labels only, with no empirical interpretation until a separate operationalization and validation record exists.

With blocks `A_i = {(i,0),(i,1)}` and `R_j = {(0,j),(1,j)}`, the generating families at each context `V` are

```
K_A(V) = { V ∩ A_0 -> V,  V ∩ A_1 -> V },
K_R(V) = { V ∩ R_0 -> V,  V ∩ R_1 -> V },
```

with the empty family covering the empty context. Unlike the legacy clauses, the global pieces are *proper* subcontexts: neither family contains an identity. Pullback along `U -> V` is intersection with `U`, so each basis is stable under pullback; refining a partition piece by the same partition returns that piece plus empty pieces.

2.2 Marks as parallel audit data

An ordinary Grothendieck topology retains generated sieves, not why a family was proposed. The reference implementation therefore maintains two distinct representations. The topology consumes the unmarked family; governance code consumes an immutable `MarkedCover`:

```
MARKED_COVER = {
  derivation_kind,          # GENERATOR | PULLBACK | COMMON_REFINEMENT
  target,
  exact_member_arrows,
  COVER_MARK_RECORD,        # cover kind, source authority, assumptions,
                            # evidence refs, review state, provenance chain
  parent_cover_sha256s
}
```

Canonical JSON and SHA-256 bind the target, members, mark, and sorted parent digests. Pullback creates a new record whose provenance chain contains its parent digest; the ACBE/RPM common refinement is a derivation node with both generating-cover digests as parents. This matters at singleton contexts, where the bare ACBE and RPM families coincide but their marked records and digests remain distinct. Deserialization of a derived record requires a parent resolver: every parent digest is rehashed, the acyclic derivation is checked, and child members and mark are recomputed; a join without exactly one ACBE and one RPM generating parent fails closed. Unknown or duplicate JSON keys and uncontrolled review states are rejected.

The mark mechanism is an external serialization and audit invariant — not a claim that `MarkedCover` is an internal topos object, a colored-site construction, or a categorical semantics of evidence. Conflating the two interfaces would repeat the original information-loss error.

3. The ambient modal candidate

The successor is *not* defined as the already-joined sheaf category — that would make the ACBE and RPM reflectors identity on every admitted object and erase the modal structure a second time. Instead, begin with the ambient presheaf topos and three topology refinements:

```
E_amb = PSh(C × L)
J_res = J_obs ∨ J_A ∨ J_R,      E_res = Sh(C × L, J_res)
M_obs = i_obs a_obs,   M_A = i_A a_A,   M_R = i_R a_R
```

Assuming `C` is small in the chosen universe:

```
NoeTop_2  := E_amb
NoeSystem_2 = (NoeTop_2, M_obs, M_A, M_R, E_res, EXTERNAL_MARK_RECORD)
```

`NoeSystem_2` is a structured package, not itself a topos or category; the provenance record is external audit data. Raw and partially resolved evidence lives in the ambient topos; only jointly resolved objects enter `E_res`. On `E_res`, both `M_A` and `M_R` restrict to identity — no text may describe them as nontrivial endomodalities of the resolved subtopos.

The v0.1 executable model covers only the L coordinate and its mark derivations. Typed observational covers in C, general arrows of C × L, and the product-site comparison remain OPEN.

4. Theorems on the finite context factor

NT2-1 (Topologies). Each of K_A, K_R is jointly exhaustive, stable under pullback, and stable under its own composition; the generated covering sieves form Grothendieck topologies J_A, J_R. *Proof:* pullbacks in L are intersections; the standard pretopology-to-topology passage applies.

NT2-2 (Subcanonicity of the context factor). Both topologies are subcanonical on L: a section of yW on V exists iff V ⊆ W and is then unique; if both pieces of a cover map into W, so does their union. This applies *only* to L — it does not repair the legacy subcanonicity gap.

NT2-3 (Distinct, non-identity modalities). Let F_A(V) be the set of bit assignments to complete ACBE blocks contained in V, restriction forgetting missing blocks. F_A satisfies ACBE gluing; at V = Omega it has four sections, while each RPM half contains no complete ACBE block, so the RPM gluing map is not injective. Thus F_A is a J_A-sheaf but not a J_R-sheaf; the symmetric F_R reverses the roles. The reflective subcategories and reflectors therefore differ, and neither reflector is the identity on PSh(L, Set).

NT2-4 (Joined topology). A_i ∩ R_j = {(i,j)}, so the common refinement of the two global partitions is the four-singleton cover, and

```
J_A ∨ J_R = J_singleton
```

— the topology generated by the singleton partition (the local label J_atom does not assert the standard category-theoretic atomic topology).

Modality composition. With explicit reflectors (R_P F)(V) = ∏_{B ∈ P, V ∩ B ≠ ∅} F(V ∩ B) (empty product a singleton), the registered internal proof verifies the universal property directly and obtains

```
M_A M_R ≅ M_atom ≅ M_R M_A     naturally on PSh(L, Set),
```

by flattening nested transverse products to atomic factors. These are natural isomorphisms, not strict equalities. The corrective narrative point: the finite modalities are *distinct but commuting*. Distinctness does not imply noncommutation, and no current artifact licenses a noncommutation claim.

Ambient lift caveat. An earlier external-product distinctness witness for PSh(C × L) is false in general (the null cover over (c, ∅) obstructs it); ambient distinctness remains OPEN, with the valid result limited to PSh(L, Set).

Executable corroboration

A dependency-free checker exhausts all 16 contexts, all 16 representables, both witness presheaves, and every generating-cover pullback (FINITE_MODEL_CERTIFICATE_v0.1.json). The exact-topology implementation enumerates all 298 sieves over the sixteen contexts and reports 168 covering sieves for both the join and singleton presentations; the modality verifier covers 18 fixtures and 3,012 restriction squares. At the bound commit the finite-site suite (8 tests), the modality-composition suite (6 tests), and the pinned Lean Gate 1 verifier all pass. Lean scope covers four-atom exhaustiveness, proper global pieces, transverse refinement, and witness separation — not the sheaf conditions, topology generation, reflector universal properties, or product-site comparison. Execution is corroboration of finite proofs, not formal verification or independent review.

5. Gluing boundary for probabilistic data

For Set-valued presheaves, gluing is bijectivity of the canonical map from global to matching local sections. Probability distributions do not inherit this by analogy: pairwise-compatible marginals may lack a global joint distribution. A probabilistic successor must pass one of four explicit gates — an acyclic/junction-tree cover with a proved extension theorem; a finite linear-program feasibility certificate; an obstruction calculation with a stated coefficient system; or a quantitative relaxation with reported residual and uncertainty. No passing Set-valued sheaf test licenses a probabilistic consistency claim.

6. Backward compatibility as a conjectural comparison

The legacy topos is preserved verbatim as `E_legacy = Sh(C, J_legacy)`. The target comparison is `E_res ≅ E_legacy^4` after atomic resolution, with a proposed fully faithful neutral-channel embedding `X |-> (X, 1, 1, 1)` and `ev_00 ∘ ι ≅ Id`. A categorical natural isomorphism is not a bit-for-bit serialization receipt: identifier and canonical-payload-hash preservation is a separate strict-equality requirement. Both round trips remain `PROOF_REQUIRED`; this paper claims no backward compatibility.

Even if proved, `E_legacy^4` exposes a strict expressivity boundary: four independent legacy channels with no remaining overlap, transition law, or cross-atom coupling. The sixteen-context construction is a finite separation and provenance reference model — not yet a substantively rich final semantics. A final successor must either name the ambient modal package as principal semantics or add typed overlaps/transition data and prove their coherence; which route is canonical remains `OPEN`, and v0.1 licenses neither.

7. Admission gates

Noetic Topos 2.0 may leave Ring 4 only after: (1) independent category-theory review of NT2-1–NT2-4 and the legacy reduction; (2) proof-assistant artifacts for the finite site, witness presheaves, and product-site comparison; (3) a verified legacy round-trip with stable identifiers and hashes; (4) a theorem-ledger merge linking every statement to exact sources; (5) proof and review of the product-site lift of modality composition and its interaction with observational coverage; (6) separate probability-gluing theorems or feasibility certificates; (7) a canonical decision between ambient semantics and genuine cross-context coupling; and (8) explicit confirmation that no empirical or product wording was licensed.

8. Conclusion

The correction and the successor fit together cleanly: the legacy ACBE/RPM clauses collapse because identities enter their sieves, and nontriviality is recovered only by changing the context generators. In the finite model the new modalities are distinct and non-identity, but they commute and jointly resolve to four atomic channels, while hash-chained mark records preserve the derivational provenance ordinary topology generation forgets. That is a coherent mathematical reference point, not a finished semantics of contextual cognition — and maintaining that distinction is itself part of the contribution.

References

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